CARTIER DUALITY AND THE LAUMON-ROTHSTEIN FOURIER TRANSFORM

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Let A be an abelian variety over a field k of characteristic 0, and let $B \cong A^\vee$ be the dual abelian scheme over k. In this short note, we explain how the the Laumon-Rothstein Fourier equivalence

$$\mathrm{D}(\mathscr{D}_A\text{-mod}) \cong \mathrm{D}(\mathrm{QCoh}(B^\natural))$$

between quasi-coherent $\mathscr{D}_A$-modules and quasi-coherent $\mathscr{O}_{B^\natural}$-modules is a manifestation of the duality between the de Rham space A_{dR} and the universal vector extension $B^\natural$. I thank Tony Pantev for explaining several key parts of this to me.

1.1. **Preliminaries.** We will define the category AlgStk to be the category of fppf sheaves of k-algebras taking values in groupoids,

$$\mathsf{AlgStk} := \mathrm{Shv}_{\mathrm{fppf}}(\mathsf{Alg}_k, \mathsf{Grpd})$$

The category of commutative k-group stacks is defined to be the subcategory $\mathsf{CGrp} \subseteq \mathsf{AlgStk}$ consisting of commutative group objects. We can define the following duality functor on commutative k-group stacks; let $\mathfrak{D} : \mathsf{CGrp}^{op} \to \mathsf{CGrp}$ be the functor

$$\mathfrak{D}(\mathcal{G}) := \underline{\mathrm{Map}}_{\mathsf{CGrp}}(\mathcal{G}, B\mathbb{G}_m)$$

Given a commutative k-group stack $\mathcal{G}$, we call $\mathfrak{D}(\mathcal{G})$ the *Cartier dual* to $\mathcal{G}$.

Example 1.1. For an abelian variety A, $\mathfrak{D}(A) \cong A^\vee$, the dual abelian variety. If $\mathbb{Z}$ is the constant group scheme, then $\mathfrak{D}(\mathbb{Z}) \cong B\mathbb{G}_m$ and $\mathfrak{D}(B\mathbb{G}_m) \cong \mathbb{Z}$. This is a 1-shifted or stacky analog of classical Cartier duality for group schemes.

The Cartier duality functor induces a canonical Fourier-Mukai transform. The universal evaluation morphism $\mathrm{ev} : \mathcal{G} \times \mathfrak{D}(\mathcal{G}) \to B\mathbb{G}_m$ is equivalent to a $\mathbb{G}_m$-torsor, and hence a line bundle $\mathscr{P}$ on $\mathcal{G} \times \mathfrak{D}(\mathcal{G})$, the *Poincaré sheaf*. This induces an integral transform

$$\Phi : \mathrm{D}_{qc}(\mathcal{G}) \to \mathrm{D}_{qc}(\mathfrak{D}(\mathcal{G}))$$

defined by $\mathscr{F} \mapsto R\pi_{2*}(\mathscr{P} \otimes^L \pi_1^* \mathscr{F})$ where the maps π_i are the projections

$$\begin{tikzcd} & \mathcal{G} \times \mathfrak{D}(\mathcal{G}) \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ \mathcal{G} & & \mathfrak{D}(\mathcal{G}) \end{tikzcd}$$

Under reasonable assumptions, this is an equivalence of categories. To use the terminology of [BB06], we say that $\mathcal{G}$ is *very nice* if smooth locally, $\mathcal{G}$ is isomorphic to (i) a finite product of abelian varieties over k, (ii) finitely generated abelian groups, or (iii) and copies of $B\mathbb{G}_m$. The authors prove:

Theorem 1.2 (2.7 of [BB06]). *If $\mathcal{G}$ and $\mathfrak{D}(\mathcal{G})$ are* very nice*, then Φ is an equivalence.*

See also [Ari04], A.1.

1.2. **The Laumon-Rothstein Fourier transform.** Let A be an abelian variety over k and $B = \mathfrak{D}(A)$ be the dual abelian variety. Recall that the *universal vector extension* is a smooth, commutative k-group scheme $B^\natural$ which is an extension

$$1 \to \mathbb{V}(\omega_A) \to B^\natural \to B \to 1 \tag{$*$}$$

where $\mathbb{V}(\omega_A)$ is the vector group of invariant differentials $\omega_A = H^0(A, \Omega^1_A)$. It's an observation due to Grothendieck that we may characterize $B^\natural$ in terms of *crystals* on A. In particular, $B^\natural$ represents the fppf sheaf $\mathrm{Sch}_k \to \mathrm{Set}$ given by

$$T \mapsto \{\text{isomorphism classes of degree 0 invertible crystals on } A \times T\},$$

[MM74]. Equivalently, $B^\natural \simeq \mathrm{Pic}^\natural(A)$ is the moduli of degree-0 line bundles on A equipped with an integrable connection.

On the other hand, the de Rham space of A is the fppf sheaf with the property

$$\mathrm{QCoh}(A_{\mathrm{dR}}) \cong \mathscr{D}_A\text{-mod}$$

In particular, this equivalence restricts to an isomorphism $\mathrm{Pic}^0(A_{\mathrm{dR}}) \cong \mathrm{Pic}^\natural(A)$. Consequently, $B^\natural$ is Cartier dual to the de Rham space; there is a natural isomorphism $B^\natural \cong \mathfrak{D}(A_{\mathrm{dR}})$. As a consequence, the abstract Fourier-Mukai equivalence for dual commutative group stacks recovers Laumon-Rothstein Fourier equivalence

$$\mathrm{D}_{qc}(A_{\mathrm{dR}}) \cong \mathrm{D}_{qc}(B^\natural)$$

There's an even more refined statement we can make: A_{dR} fits into an exact sequence

$$1 \to \widehat{A} \to A \to A_{\mathrm{dR}} \to 1 \tag{$\dagger$}$$

where $\widehat{A}$ denotes the formal completion of A along the identity section. Applying the duality functor $\mathfrak{D}$ to ($\dagger$), we get an exact sequence

$$\to \underline{\mathrm{Map}}_{\mathsf{CGrp}}(\widehat{A}, \mathbb{G}_m) \to \mathfrak{D}(A_{\mathrm{dR}}) \to \mathfrak{D}(A) \to \mathfrak{D}(\widehat{A}) \to$$

analogous to the long exact sequence in cohomology with coefficients in $\mathbb{G}_m$.

Lemma 1.3. *We have a natural isomorphism*

$$\underline{\mathrm{Map}}_{\mathsf{CGrp}}(\widehat{A}, \mathbb{G}_m) \cong \mathbb{V}(\omega_A)$$

Proof. Since k has characteristic 0,

$$\mathrm{Hom}_{grp}(\widehat{A}, \mathbb{G}_m) \cong \mathrm{Hom}_k(\mathrm{Lie}(\widehat{A}), \mathrm{Lie}(\mathbb{G}_m)) \cong \mathrm{Hom}_k(\mathrm{Lie}(A), k)$$

and since $e^*\Omega^1_{A/k} \cong \omega_A$ (see 2.2ii), it follows that $\mathrm{Lie}(A)^\vee = \omega_A$. $\square$

Consequently, exact sequences $(*)$ and $(\dagger)$ are dual to each other up to a shift!

Remark 1.4. We close with an observation about the relation with the Hodge filtration on $H^1_{\mathrm{dR}}(A/k)$. By a standard result of deformation theory,

$$\mathrm{Lie}(\mathfrak{D}(X)) = H^1(X, \mathscr{O}_X)$$

Thus if we apply the Lie algebra functor to the exact sequence $(*)$, we obtain

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Lie}(\mathbb{V}(\omega_A)) & \longrightarrow & \mathrm{Lie}(B^\natural) & \longrightarrow & \mathrm{Lie}(B) & \longrightarrow & 0 \\ & & \| & & \| & & \| & & \\ & & H^0(A, \Omega^1_{A/k}) & \longrightarrow & H^1(A_{\mathrm{dR}}, \mathscr{O}_{A_{\mathrm{dR}}}) & \longrightarrow & H^1(A, \mathscr{O}_A) & & \end{array}$$

which identifies with the Hodge filtration on $H^1_{\mathrm{dR}}(A/k)$.

Remark 1.5. After writing this short exposé, I became aware that this entire story (and much more) is already contained in the thesis of Gabriel Riberio [Rib24], who provided some very insightful comments. In particular, remark 2.3.24 provides a proof of the duality between the sequences $(*)$ and $(\dagger)$.

References

[Ari04] D. Arinkin. Appendix to [DP04], 2004.

[BB06] R. Bezrukavnikov and A. Braverman. Geometric Langlands correspondence for D-modules in prime characteristic: the GL(n) case, 2006.

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[MM74] B. Mazur and W. Messing. *Universal Extensions and One Dimensional Crystalline Cohomology*, volume 370 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin-Heidelberg, 1974.

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